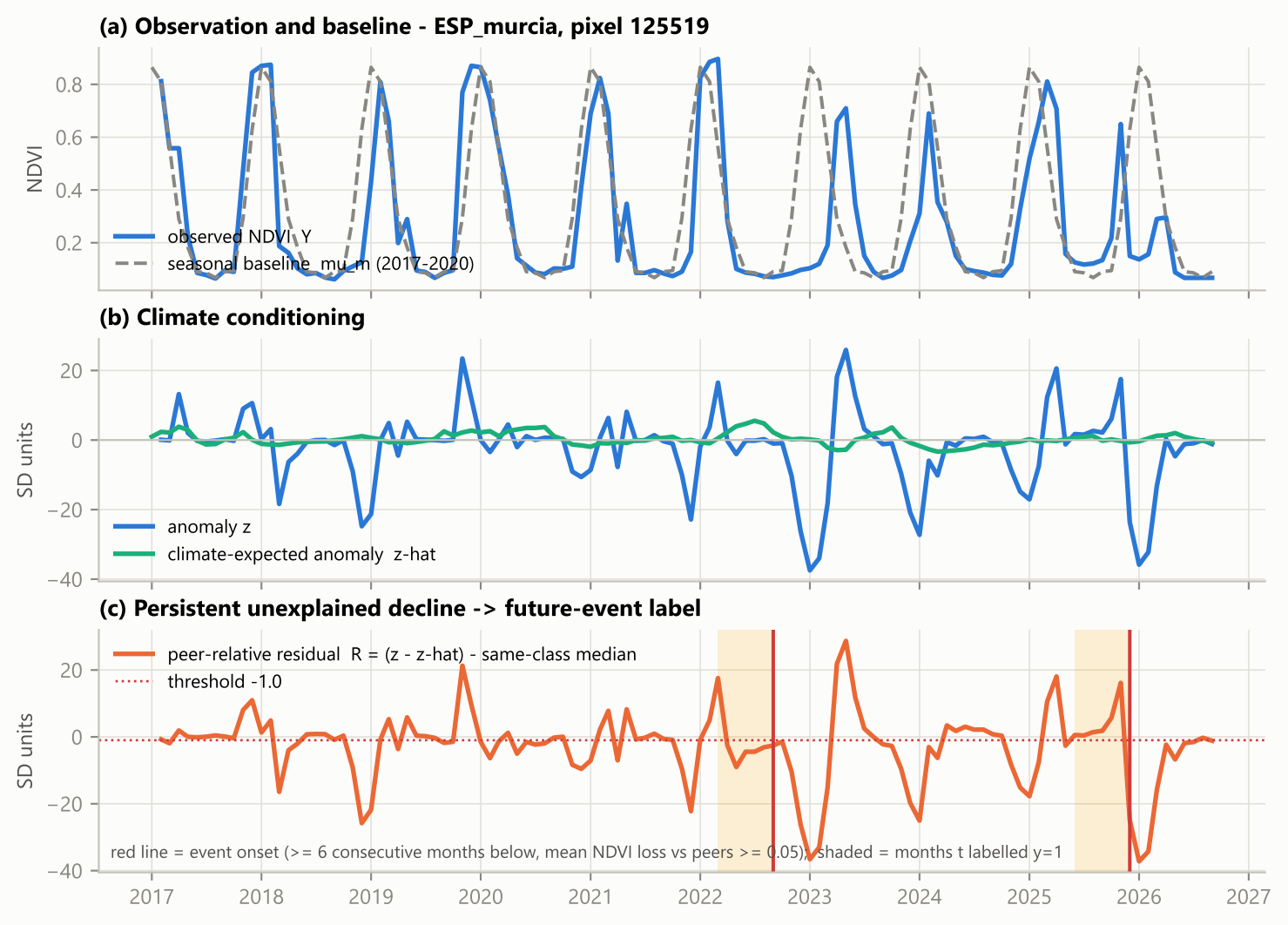
(a) Observation and baseline - ESP_murcia, pixel 125519
NDVI
0.8
0.6
0.4
0.2
observed NDVI_Y
seasonal baseline_mu_m (2017-2020)
(b) Climate conditioning
SD units
20
0
−20
−40
anomaly z
climate-expected anomaly z-hat
(c) Persistent unexplained decline -> future-event label
SD units
peer-relative residual R = (z - z-hat) - same-class median
threshold -1.0
red line = event onset (>= 6 consecutive months below, mean NDVI loss vs peers >= 0.05); shaded = months t labelled y=1
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027